

DELOITTE AUSTRALIA

Technology Virtual Experience Program

Forage Simulation – Project Report

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Program Overview

Program	Deloitte Australia – Technology Virtual Experience
Platform	Forage
Client	Daikibo Industrials
Tasks Completed	Task 1: Data Analysis & Visualization Task 2: Forensic Technology
Tools Used	Tableau, Microsoft Excel

This report summarizes the work completed during the Deloitte Australia Technology Virtual Experience simulation hosted on Forage. The simulation involved two analytical tasks for a fictional industrial client, Daikibo Industrials, covering telemetry data analysis and gender pay equality classification.

Task 1: Data Analysis & Visualization

Background

Daikibo Industrials operates 4 global factories, each equipped with 9 types of machines that transmit telemetry data every 10 minutes. The client collected one month of data (May 2021) and wanted answers to two key business questions:

- In which factory location did machines break down most frequently?
- Which machine types broke down most often in that highest-downtime location?

Factories Analysed

Daikibo Factory Meiyō Tokyo, Japan	Daikibo Factory Seiko Osaka, Japan	Daikibo Berlin Berlin, Germany	Daikibo Shenzhen Shenzhen, China
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Methodology

The telemetry data was imported into Tableau from a unified JSON file. The following steps were performed:

- A calculated measure field "Unhealthy" was created, assigning a value of 10 for every unhealthy machine status (representing 10 minutes of potential downtime per message).
- A bar chart titled "Down Time per Factory" was built to compare total downtime across all 4 locations.
- A second bar chart titled "Down Time per Device Type" was created on a separate sheet.
- A Dashboard was assembled combining both charts, with the factory chart configured as a filter — selecting a factory dynamically updates the device type chart to show only that factory's data.

Key Finding

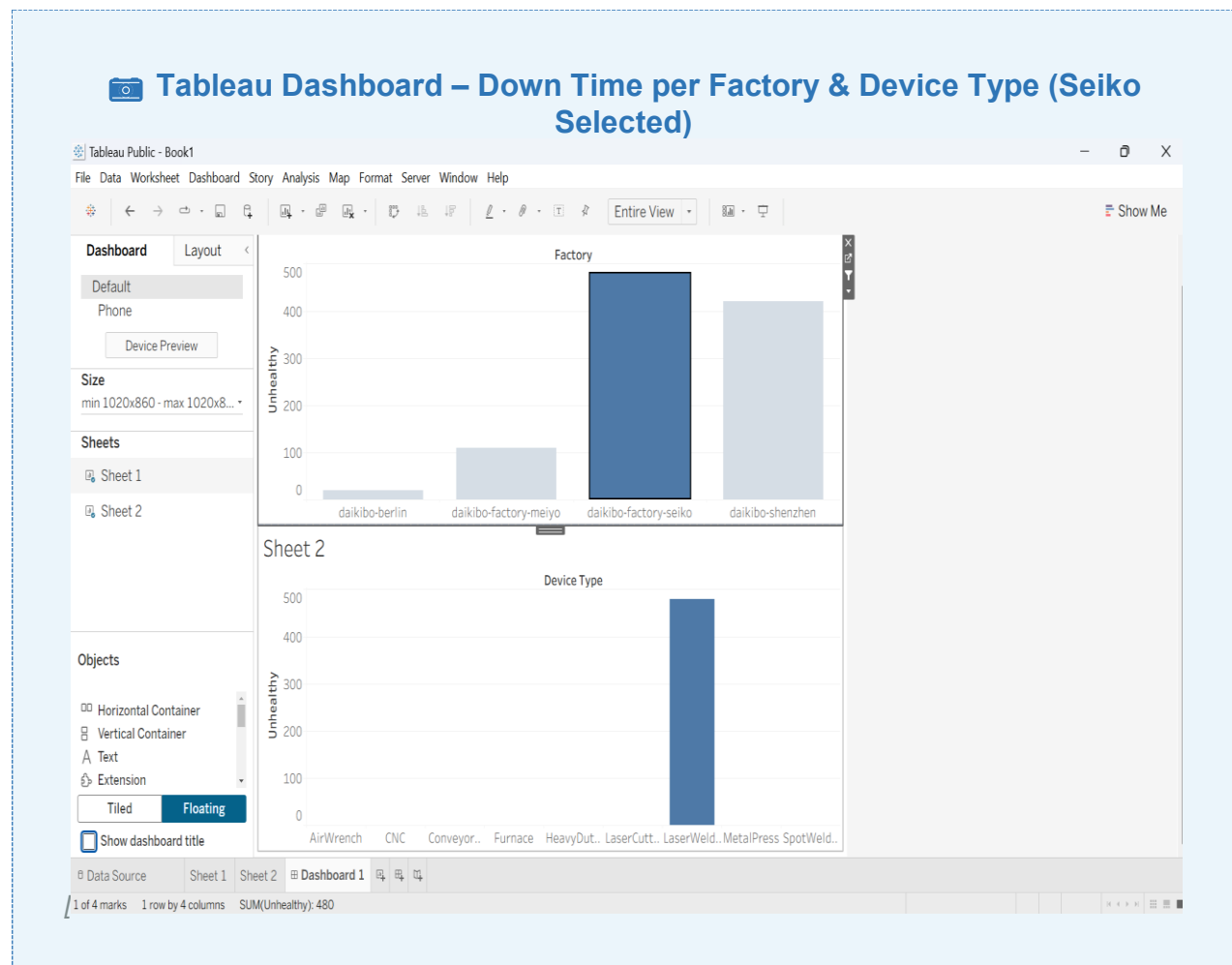


Highest Downtime Location: Daikibo Factory Seiko (Osaka, Japan)

Total Unhealthy Readings: 480

Tableau Dashboard Screenshot

The dashboard below shows the Down Time per Factory chart with Daikibo Factory Seiko selected, filtering the device-type breakdown to display only Seiko's machine downtime data.



Task 2: Forensic Technology – Gender Pay Equality

Background

Following internal complaints about gender pay inequality, Daikibo Industrials engaged Deloitte's Forensic Technology team. An algorithm was developed to quantify gender pay equality across all job roles and factory locations, producing an Equality Score for each combination.

Task Description

The provided Excel file (Equality Table.xlsx) contained 3 columns: Factory, Job Role, and Equality Score (an integer ranging from -100 to +100, where 0 is ideal). The task required adding a 4th column — Equality Class — to categorize each score using conditional logic.

Classification Logic

Equality Class	Score Range	Example
Fair	-10 to +10 (inclusive)	Score: 10 → Fair
Unfair	< -10 or > 10 (outside ±10 but within ±20)	Score: -9 → Unfair
Highly Discriminative	< -20 or > 20	Score: -30 → Highly Discriminative

Excel Formula Used

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=IF (ABS (C2) <=10, "Fair", IF (ABS (C2) <=20, "Unfair", "Highly Discriminative"))
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This formula was applied to column D (Equality Class) across all rows of the dataset. It uses ABS() to handle both positive and negative scores symmetrically, and nested IF logic to assign the correct classification tier.

Output

The completed Excel file with the added Equality Class column was submitted as the Task 2 deliverable. Each row now contains a classification (Fair, Unfair, or Highly Discriminative) alongside the factory, job role, and equality score.

Summary & Key Takeaways

Task 1: Data Visualization	Task 2: Forensic Technology
• Imported JSON telemetry data into Tableau • Created Unhealthy calculated field (value = 10) • Built two bar charts and a linked dashboard • Identified Seiko as highest downtime factory (480 readings)	• Analysed gender pay equality dataset • Designed 3-tier classification logic • Applied Excel formula using ABS() + nested IF • Delivered classified Equality Table

This virtual experience reinforced practical skills in data visualization, dashboard design, and business-oriented analytical thinking — directly applicable to the Analyst role at Deloitte's GPS SSC team.